

OS & SHELL LEARNING

>_ MY SHELL

Interactive Linux Learning Environment

A Beginner-Centric Shell for Learning Linux and OS Concepts

C · Linux / Ubuntu

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ONE ENVIRONMENT

>_ What is MyShell?

Linux shell execution, OS explanations, guided practice, quiz-based assessment, instant scoring and progress tracking.

1 LEARN →	2 UNDERSTAND →	3 EXECUTE →
4 PRACTICE →	5 QUIZ →	6 TRACK

“The shell itself becomes the classroom.”

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>_ Accuracy of Results

```
--- QUIZ ---
What does ls do?
A) Lists files and directories
B) Creates a new file
C) Removes a file

Your answer: A
Score: 1/1 √
Progress: ls [Mastered √]
```

Real Linux command execution; explanations tied to the command being practiced.

Immediate quiz evaluation, correct/incorrect feedback, score calculation, visible learning status, and progress tracking.

Accuracy shown through deterministic command behavior and automated quiz evaluation; no percentage claimed.

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>_ Technical Competence

USER → COMMAND INPUT → COMMAND PARSING

```
fork() → PROCESS CREATION
```

```
exec() → COMMAND EXECUTION
```

```
wait() → PROCESS SYNCHRONIZATION → OUTPUT
```

Pipes · I/O Redirection · Process Management
Learning Module · Quiz Module · Progress Persistence

C | Linux/Ubuntu | GCC | POSIX | JSON

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LEARNING WORKFLOW

>_ From Command to Learning

1

Select Lesson

ls →

2

Understand Command

Explain it →

3

Execute Command

$ ls →

4

Practice

Try it →

5

Take Quiz

Answer →

6

Receive Score

Score 1/1 →

7

Update Progress

Mastered

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>_ Innovation: Integration, Not Reinvention

Traditional

Documentation → Terminal
→ Separate Practice
→ Separate Quiz
→ Manual Progress

Learning spans separate tools.

MyShell

Learn → Understand → Execute
→ Practice → Quiz → Score
→ Track Progress

OS concepts beside commands; automated assessment and persistent progress, without switching tools.

“The innovation lies in integration, not reinvention.”

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CRITERION 3 · 10 MARKS

>_ Sustainable & Extensible Design

Lightweight today

- C-based; Linux/Ubuntu
- GCC compilation
- Minimal resource requirements
- Text-based interface
- JSON progress storage

Extensible tomorrow

Additional Linux commands, more learning modules and expanded practice.

Adaptive difficulty, personalized exercises, weak-area detection, personalized learning paths and adaptive recommendations.

Future extension opportunities

CURRENT SYSTEM → EXTENSIBLE LEARNING PLATFORM

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CRITERION 4 · 10 MARKS

>_ Utility to Society

Students

Beginner-friendly Linux learning.

Guided practice, immediate feedback and self-paced study build better OS understanding.

Educators

A structured teaching tool.

Classroom demonstrations combine practice, assessment and progress visibility.

Institutions

Lightweight and Linux-based.

Fits OS/Linux courses, supports independent learning and extends with new commands and lessons.

“Making Linux learning accessible, guided and measurable.”

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END-TO-END WALKTHROUGH

>_ Live Demonstration

```
MyShell> 1
LESSON: ls
Lists files and directories; use it to inspect a directory.
Example: $ ls
inside Linux: fork() → exec(ls) → wait()
PRACTICE: try typing ls
QUIZ: What does ls do? Answer A

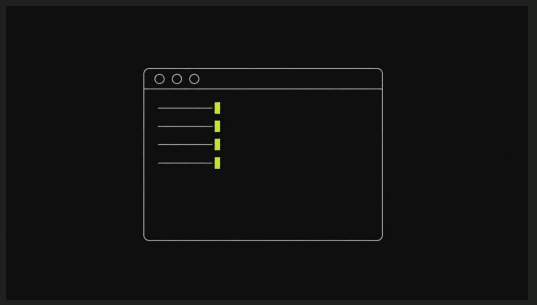
Score: 1/1 √
PROGRESS: ls [Mastered √]
pwd [In Progress...] · cd [Not Started]
```

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FUNCTIONAL EVIDENCE

>_ Evidence of Implementation

- Linux commands execute through the shell; fork(), exec() and wait() demonstrate process handling; pipes chain commands such as ls | grep.
- Quiz answers produce scores; progress states are maintained; JSON storage preserves progress.



Command execution
→ OS-level process handling
→ Learning explanation
→ Practice → Assessment
→ Progress tracking

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